

A server rack with glowing blue lights and a bokeh background of yellow and blue circles.

Unit 6: Computer Storage

COMPUTER STORAGE

A computer can function with only processing, memory, input, and output devices. To be really useful, however, a computer also needs a place to keep program files and related data when they are not in use.

The purpose of STORAGE is to hold data permanently



Storage Devices

- A **storage device** is used in the computers to store the data.
- Another way to define, A **storage device** is any hardware capable of holding information either temporarily or permanently.
- Provides one of the core functions of the modern computer.

Why is storage needed in a computer?

- Without a storage device, a computer cannot save or remember any settings or information and would be considered a dumb terminal
- Although a computer can run with no storage device, it would only be able to view information, unless it was connected to another computer that had storage capabilities
- Even a task, such as browsing the Internet, requires information to be stored on your computer

Bits and bytes

A computer only understands the numbers **0** or **1**, or whether a switch is **on** or **off**.

We call those 1s and 0s '**bits**' – **binary digits**.

A **byte** (**made up of 8 bits**) is enough computer memory to store a single character of data (e.g. the letter F).

The computer uses a code to understand what each bit pattern means.

Using the **ASCII** code, for instance, the letter F is 70 and has a bit pattern of 01000110.



ASCII for Capital Letters				
65	A		78	N
66	B		79	O
67	C		80	P
68	D		81	Q
69	E		82	R
70	F		83	S
71	G		84	T
72	H		85	U
73	I		86	V
74	J		87	W
75	K		88	X
76	L		89	Y
77	M		90	Z

ASCII

American Standard Code for Information Interchange is a code which represents English characters as numbers. Each letter is assigned a number. For example, A = 65.

Most computers use ASCII codes. This makes it possible to transfer data from one computer to another by changing the ASCII code into a binary pattern.

BASIC UNIT OF STORAGE

1. **Bit** : Computer works with binary digits. These digits are in the form of 0's and 1's. A binary digit is called **bit**. One bit takes one storage location in memory. It is the smallest unit for data storage.
2. **Byte** : A collection of eight bits is called **byte**. It is used to store single character. The capacity of the memory or the storage is expressed in terms of bytes.
3. **Kilobyte** : A kilobyte consists of 1024 bytes. It is denoted by **KB**.
4. **Megabyte** : megabyte consists of 1024 kilobytes. It is denoted by **MB**.
5. **Gigabyte**: A gigabyte consists of 1024 megabytes. It is denoted by **GB**.
6. **Terabyte**: A terabyte consists of 1024 gigabytes. It is denoted by **TB**

Data storage

The amount of **data stored** is measured in **kilobytes (KB)**.

Confusingly, **1KB** is actually 1,024 bytes (2^{10}), not 1,000 as you might expect, but most people think in multiples of 1,000.

1 bytes= 8bits

1megabyte (MB) =1,024 KB (2^{20})

1 gigabyte (GB) =1,024 MB (2^{30})

1 terabyte (TB) = 1,024GB (2^{40}).

- **1. Purpose of Memory**

- Memory is critical for the operation of any computing device. It serves several purposes:

- **Stores Data and Instructions:** Retains information temporarily (RAM) or permanently (ROM).
- **Enables Processing:** Provides space for data manipulation during execution.
- **Facilitates Access:** Memory types like cache and registers ensure faster access to frequently used data.

Storage Devices

Types of Storage

There are two type of storage:

- Primary Storage
- Secondary Storage

Storage Devices

Primary Storage

- Also known as **main memory**.
- Primary storage is a storage location that holds memory for short periods of times while the computer is on.
- The CPU continuously reads instructions stored there and executes them as required.
- Example:
 - RAM
 - ROM

Primary Memory (Volatile Memory):

- **Definition:** Temporary storage directly accessible by the CPU.

- **Examples:**

1. RAM (Random Access Memory):

- **Types:**

- **Static RAM (SRAM):** Faster, used in cache memory.

- **Dynamic RAM (DRAM):** Slower, used in main memory.

- **Example:** DDR4 SDRAM used in modern computers.

- **Purpose:** Stores data and instructions currently in use.

2. Cache Memory:

- Located inside or close to the CPU.

- **Example:** L1, L2, L3 caches in modern processors.

- **Purpose:** Reduces latency by holding frequently accessed data.

3. Registers:

- Smallest and fastest memory within the CPU.

- **Example:** Program Counter (PC), Accumulator.

- **Purpose:** Holds data for immediate execute

Storage Devices

Secondary Storage

- Secondary storage is a storage medium that holds information until it is deleted or overwritten regardless if the computer has power
- It is not directly accessible by the CPU.
- Computer usually uses its input/output channels to access secondary storage and transfers the desired data using intermediate area in primary storage.
- **Example:**
 - Hard disk

Storage Devices

Secondary Storage

Hard Disk



Storage Devices

Secondary Storage

Hard Disk

	Internal Hard disk	External Hard disk
Portability	No	Yes
Price	Less expensive	More expensive
Speed	Fast	Slow
Size	Big	Small

Secondary Memory (Non-Volatile Memory):

- **Definition:** Permanent storage for data and programs.

- **Examples:**

- 1. Hard Disk Drives (HDD):**

- Uses magnetic platters and read/write heads.
- Example: Seagate Barracuda 2TB.
- **Purpose:** Storing OS, software, and large files.

- 2. Solid-State Drives (SSD):**

- Uses flash memory for faster performance.
- Example: Samsung 970 EVO NVMe SSD.
- **Purpose:** Faster boot times and application performance.

- 3. Optical Discs:**

- Examples:
 - **CD-ROM:** Read-only, stores 700MB.
 - **DVD:** Stores 4.7GB to 8.5GB.
 - **Blu-ray:** Stores 25GB to 128GB.
- **Purpose:** Media storage, backups.

Types of Secondary storage device

- ❑ Optical storage device-

CD , DVD ,Blu-Ray

- ❑ Magnetic storage device-

Floppy disk, Hard disk

- ❑ Solid state device-

Flash memory & Memory stick

Tertiary Memory (Removable Storage):

- **Definition:** Data stored externally, often used for backups.

- **Examples:**

1. USB Flash Drives:

- Example: SanDisk Ultra 64GB.
- **Purpose:** Portable storage for files.

2. Memory Cards:

- Example: SD cards used in cameras and smartphones.
- **Purpose:** Extending device storage.

3. Magnetic Tape:

- Example: LTO Ultrium 9 (18TB capacity).
- **Purpose:** Long-term archival storage.

Cache Memory

- ❑ Cache memory is a very high speed semiconductor memory which can speed up the CPU.
- ❑ It acts as a buffer between the CPU and the main memory.
- ❑ It is used to hold those parts of data and program which are most frequently used by the CPU.
- ❑ The parts of data and programs are transferred from the disk to cache memory by the operating system, from where the CPU can access them.

Disadvantages

- Cache memory has limited capacity.
- It is very expensive.
- It can store data for temporary use.

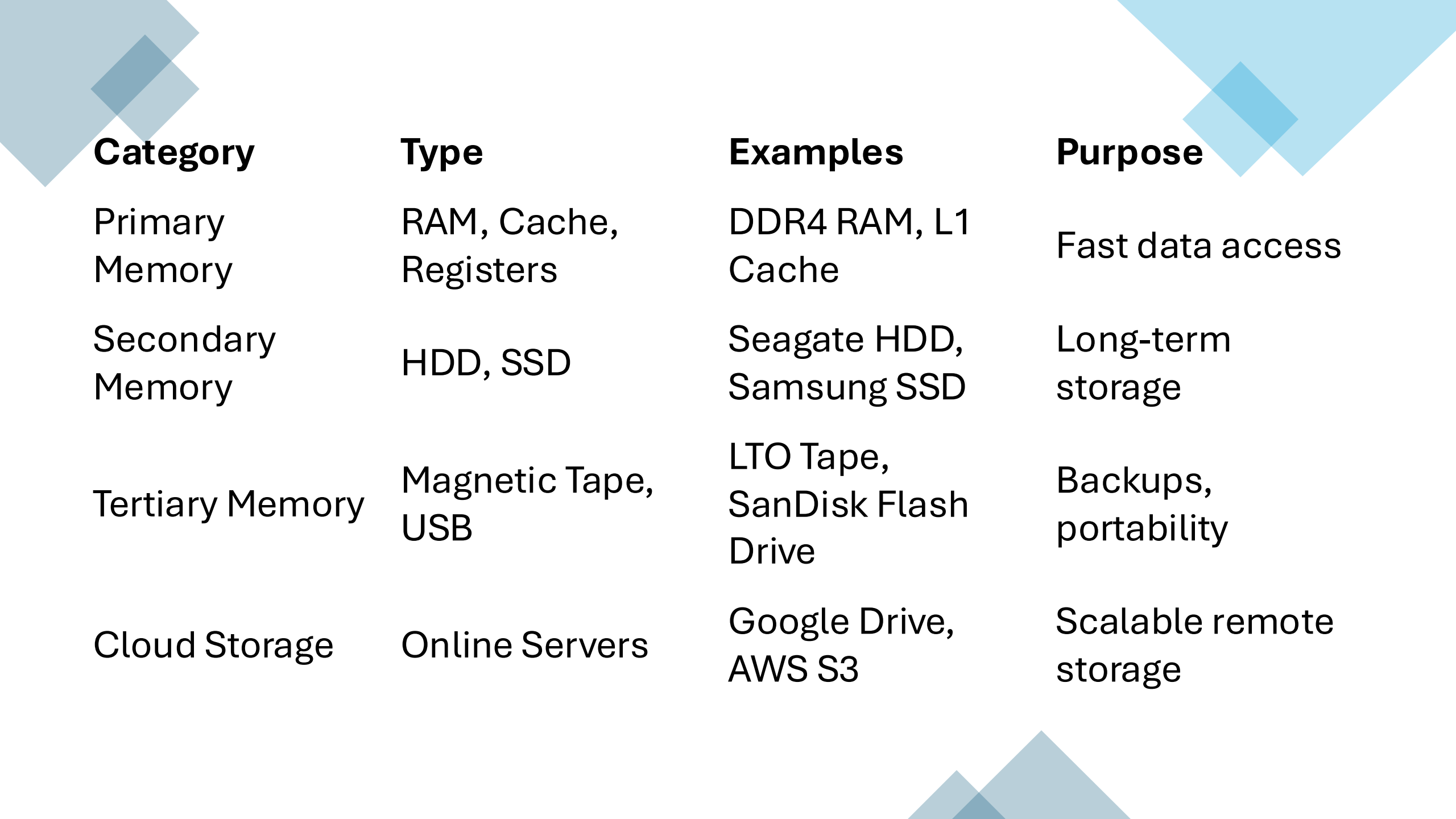
Advantage

- It is faster than the main memory.
- It consumes less excess time as compared to the main memory.



Cloud Storage:

- **Definition:** Internet-based storage where data is kept on remote servers.
- **Examples:**
 - Google Drive, Dropbox, AWS S3.
- **Purpose:** Scalable storage with remote access.



Category	Type	Examples	Purpose
Primary Memory	RAM, Cache, Registers	DDR4 RAM, L1 Cache	Fast data access
Secondary Memory	HDD, SSD	Seagate HDD, Samsung SSD	Long-term storage
Tertiary Memory	Magnetic Tape, USB	LTO Tape, SanDisk Flash Drive	Backups, portability
Cloud Storage	Online Servers	Google Drive, AWS S3	Scalable remote storage

Access Time

Access time refers to the time it takes for memory or storage to provide data to the CPU.



Components of Access Time

1. **Seek Time:**

- Time taken by the read/write head of a storage device (e.g., HDD) to move to the track where the required data is located.
- **Example:** In a hard disk drive, the mechanical arm moves to the correct position.

2. **Latency (Rotational Delay):**

- The delay caused by waiting for the desired sector of the storage medium to rotate under the read/write head.
- Common in **HDDs**, where platters spin at specific speeds (e.g., 5400 RPM or 7200 RPM).

3. Data Transfer Time:

- Time taken to transfer data from memory/storage to the CPU.
- This depends on the bandwidth of the storage interface (e.g., SATA, NVMe).

4. Total Access Time:

- Sum of seek time, latency, and data transfer time.

Access Time by Storage Type

- Access time varies significantly across storage and memory types due to differences in technology:

Storage Type	Access Time	Description
Registers	1-2 nanoseconds (ns)	Fastest, located within the CPU.
Cache Memory	10-20 nanoseconds (ns)	High-speed memory close to the CPU.
RAM (Main Memory)	50-70 nanoseconds (ns)	Temporary storage for active data and instructions.
SSD (Solid-State Drive)	0.1 milliseconds (ms)	Uses NAND flash memory, no moving parts.
HDD (Hard Disk Drive)	5-10 milliseconds (ms)	Mechanical components increase access time.
Optical Discs (CD/DVD)	~200 milliseconds (ms)	Slow due to rotational delay and laser reading mechanisms.
Magnetic Tape	Several seconds (sequential access)	Sequential access makes it slower for random data retrieval.

Factors Affecting Access Time

1. Technology:

- Faster storage technologies (e.g., SSDs, RAM) have lower access times compared to mechanical or optical storage.

2. Storage Type:

- Random-access storage (e.g., RAM) offers consistent access times for any location.
- Sequential-access storage (e.g., magnetic tape) is slower for random data access.

3. Bandwidth:

- The speed of data transfer interfaces (e.g., SATA, NVMe, USB) affects overall access time.

4. System Design:

- Proximity of memory/storage to the CPU reduces latency (e.g., cache memory is faster than main memory).

5. Workload:

- High concurrent requests or fragmented storage can increase access time.

Importance of Access Time

1.System Performance:

- Faster access time enhances the overall speed and responsiveness of the system.

2.User Experience:

- Reduces lag in applications, boot times, and file loading.

3.Application-Specific Requirements:

- Real-time applications (e.g., gaming, video editing) require storage with low access times.
- Archival storage prioritizes capacity over speed.

Improving Access Time

1.Upgrading Hardware:

- Replace HDDs with SSDs for faster storage access.
- Increase RAM capacity to reduce reliance on slower virtual memory.

2.Optimizing Software:

- Defragmentation of HDDs to reduce seek time.
- Use file systems optimized for the storage medium (e.g., NTFS for HDDs, APFS for SSDs).

3.Caching:

- Frequently accessed data is stored in faster memory (e.g., cache or RAM).

4.Parallelism:

- Use of RAID configurations or multi-channel memory to improve throughput.

Examples

1.Gaming:

- Games load faster on SSDs due to lower access times compared to HDDs.

2.Data Centers:

- Use NVMe SSDs for high-speed access to large datasets.

3.Backup Systems:

- Magnetic tape is used where access speed is less critical but a large capacity is required.

Comparison: SSD vs HDD Access Time

Feature	HDD	SSD
Access Time	5-10 milliseconds	~0.1 milliseconds
Seek Time	Significant (mechanical)	Negligible (electronic)
Latency	High due to rotation	Minimal
Performance	Slower for random access	Faster for all types of access

- **Access Time**

- **Definition:** The time taken to retrieve data from memory.

- **Examples:**

- **Registers:** 1-2 nanoseconds.
- **Cache Memory:** 10 nanoseconds.
- **RAM:** 50-70 nanoseconds.
- **Hard Drive:** 5-10 milliseconds.
- **Optical Disc:** 200 milliseconds.

Register

Definition:

- Registers are small, high-speed storage locations built directly into the CPU.
- They temporarily hold data and instructions that the CPU is actively using.

Characteristics:

- **Speed:** Fastest storage in the hierarchy.
- **Size:** Very small (usually a few bytes).
- **Access Time:** 1-2 nanoseconds.
- **Volatility:** Volatile (data is lost when the power is off).

Functions:

- Store intermediate results during computations.
- Hold the instruction currently being executed.
- Maintain essential CPU control data (e.g., Program Counter).

Examples:

- **Accumulator Register:** Stores the result of arithmetic and logic operations.
- **Instruction Register (IR):** Holds the current instruction being executed.
- **Program Counter (PC):** Stores the address of the next instruction to be executed.

Advantages:

- Provides the fastest data access.
- Essential for the CPU's operation.

Disadvantages:

- Limited storage capacity.
- Expensive to build.

Comparison of Storage Levels



Feature	Register	Main Memory	Backing Store
Speed	Fastest	Fast	Slow
Size	Smallest	Moderate	Largest
Access Time	1-2 nanoseconds	50-70 nanoseconds	Milliseconds
Volatility	Volatile	Volatile	Non-volatile
Purpose	Temporary data for CPU	Active data storage	Long-term data storage

Principles of Construction

Magnetic Tape Drives:

•Working:

- Stores data on a tape coated with magnetic material.
- Data is written and read sequentially.

•Examples:

- Used in archival storage, such as IBM 3592 tape drives.

•Advantages:

- Cost-effective for large data volumes.

•Disadvantages:

- Slow access times.

Magnetic Disk Drives (HDD):

•Working:

- Rotating magnetic platters store data.
- Read/write heads access data magnetically.

•Examples:

- Seagate Barracuda HDD.

•Advantages:

- High capacity, low cost per GB.

•Disadvantages:

- Slower than SSDs.

Solid-State Drives (SSD):

•Working:

- Uses NAND flash memory.
- No moving parts, making it faster and more durable.

•Examples:

- Samsung 970 EVO Plus NVMe SSD.

•Advantages:

- High speed, low power consumption.

•Disadvantages:

- Higher cost per GB.

Optical Discs:

•CD-ROM:

• Working:

- Data is stored as microscopic pits and lands.
- A laser reads these patterns to retrieve data.

• Advantages:

- Portable, durable.

• Disadvantages:

- Limited capacity (700MB).

•DVD:

- Similar to CD-ROM but with higher density.
- Example: Double-layer DVDs store up to 8.5GB.

•Blu-ray:

- Uses blue lasers for higher precision.
- Example: 50GB storage for dual-layer discs.

RAM Demonstration: Open a Heavy Application

Objective:

To demonstrate how increasing the amount of **Random Access Memory (RAM)** can speed up application performance and multitasking.

Steps:

1.Setup:

1. Use two computers (or the same computer with different RAM capacities).
 1. Example: One system with 4GB RAM and another with 16GB RAM.
2. Install a resource-intensive application (e.g., Adobe Photoshop, a video editing tool, or a large dataset in Excel).

2.Demonstration:

1. Open the heavy application on both systems.
2. Monitor the time it takes to load and respond to commands.
3. On the 4GB RAM system:
 1. Observe lag or slower responsiveness, especially with multiple tabs or applications open.
4. On the 16GB RAM system:
 1. Notice faster load times and smoother multitasking.

3.Analysis:

1. Explain that **RAM** temporarily stores active data and instructions, reducing the need to repeatedly access slower secondary storage (like HDDs or SSDs).

Highlight how insufficient RAM leads to system reliance on virtual memory, slowing down performance.

Outcome:

More RAM allows the system to handle larger workloads efficiently, improving performance for heavy applications and multitasking.

2. HDD vs SSD: Compare Boot Times

Objective:

To showcase the difference in boot times between laptops with **Hard Disk Drives (HDDs)** and **Solid-State Drives (SSDs)**.

Steps:

1.Setup:

1. Use two laptops:
 1. One with an HDD (e.g., 1TB 5400 RPM HDD).
 2. One with an SSD (e.g., 500GB NVMe SSD).
2. Ensure both laptops have the same operating system installed (e.g., Windows 11).

2.Demonstration:

1. Turn off both laptops.
2. Start both laptops simultaneously and measure the time they take to boot up (from pressing the power button to reaching the login screen).
3. Observe:
 1. The HDD laptop may take 30-60 seconds to boot.
 2. The SSD laptop typically boots in 10-15 seconds.

3.Analysis:

1. Explain that **SSDs** use flash memory and have no moving parts, enabling faster data retrieval.
2. **HDDs**, with spinning platters and mechanical read/write heads, are slower due to physical limitations.

Outcome:

SSDs significantly improve boot times and overall system performance, especially for resource-intensive tasks.

3. Cloud Storage: Use Google Drive to Upload and Access Files

Objective:

To demonstrate the convenience and flexibility of **cloud storage** for file sharing and accessibility.

Steps:

1.Setup:

1. Use any device (computer, smartphone, or tablet) with an internet connection.
2. Create or log into a Google Drive account.

2.Demonstration:

1. Step 1: Upload a file to Google Drive.

1. Example: Upload a 50MB PDF or a presentation.
2. Observe the upload speed based on internet bandwidth.

2. Step 2: Access the file on a different device.

1. Log in to the same Google Drive account on another device (e.g., smartphone).
2. Open or download the file from Google Drive.

3. Step 3: Share the file.

1. Use the "Share" option in Google Drive to send the file to another user via email or a link.

3.Analysis:

1. Highlight the **advantages** of cloud storage:

1. Accessibility from any internet-connected device.
2. Real-time collaboration on files (e.g., Google Docs/Sheets).
3. Automatic backups and version control.

2. Mention potential **disadvantages** like reliance on internet access and data security concerns.

Outcome:

Cloud storage simplifies file management, enhances collaboration, and provides flexibility for accessing files across devices.

Summary Table



Demonstration	Key Focus	Outcome
RAM Demonstration	Increased performance with more RAM	Faster application loading and smoother multitasking.
HDD vs SSD	Storage type impact on boot times	SSDs significantly reduce boot times compared to HDDs.
Cloud Storage	File accessibility and sharing	Seamless access, sharing, and backup of files from multiple devices.